Role

Column	Data Type	Constraints	Purpose
id	BIGINT	Primary Key	Unique identifier
role_name	VARCHAR(255)	UNIQUE	Name of the role
description	TEXT	Nullable	Description of the role
created_at	TIMESTAMP	Auto	Created timestamp
updated_at	TIMESTAMP	Auto	Updated timestamp

User

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	User ID
role_id	BIGINT	FK	User role
first_name	VARCHAR(255)	Required	First name
last_name	VARCHAR(255)	Required	Last name
email	VARCHAR(255)	UNIQUE	Login email
email_verified_at	TIMESTAMP	Nullable	Laravel authentication
password	VARCHAR(255)	Required	Hashed password
profile_picture	VARCHAR(255)	Nullable	Image path
remember_token	VARCHAR	Laravel	Remember Me
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Category

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Category ID
category_name	VARCHAR(255)	UNIQUE	Category name
description	TEXT	Nullable	Category description
created_at	TIMESTAMP	Auto	Laravel timestamp
updated_at	TIMESTAMP	Auto	Laravel timestamp

Course

Column	Data Type	Constraints
id	BIGINT	PK
mentor_id	FK	Restrict
category_id	FK	Restrict
title	VARCHAR(255)	Required
slug	VARCHAR(255)	Unique
description	TEXT	Required
thumbnail	VARCHAR(255)	Nullable
status	ENUM	Draft / Published / Archived
created_at	TIMESTAMP	Auto
updated_at	TIMESTAMP	Auto

Lesson

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Lesson ID
course_id	BIGINT	FK → courses.id	Parent course
title	VARCHAR(255)	Required	Lesson title
content	LONGTEXT	Required	Lesson content (HTML)
order_number	INTEGER	Required	Lesson order
estimated_duration	INTEGER	Nullable	Duration in minutes
status	ENUM	Draft / Published	Lesson visibility
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Assessment

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Assessment ID
lesson_id	BIGINT	FK → lessons.id	Parent lesson
title	VARCHAR(255)	Required	Assessment title
description	TEXT	Nullable	Instructions
passing_score	INTEGER	Required	Passing percentage
time_limit	INTEGER	Nullable	Time limit (minutes), NULL = no limit
max_attempts	INTEGER	Nullable	NULL = unlimited attempts
status	ENUM	Draft / Published	Assessment status
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Question

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Question ID
assessment_id	BIGINT	FK → assessments.id	Parent assessment
question_text	TEXT	Required	The question
question_type	ENUM	Required	MCQ, TrueFalse, ShortAnswer
option_a	TEXT	Nullable	MCQ Option A
option_b	TEXT	Nullable	MCQ Option B
option_c	TEXT	Nullable	MCQ Option C
option_d	TEXT	Nullable	MCQ Option D
correct_answer	TEXT	Required	Correct answer
explanation	TEXT	Nullable	Explanation shown after answering
marks	INTEGER	Default 1	Marks awarded
order_number	INTEGER	Required	Display order
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Enrollment

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Enrollment ID
student_id	BIGINT	FK → users.id	Student
course_id	BIGINT	FK → courses.id	Enrolled course
status	ENUM	Active, Completed, Dropped	Current enrollment status
enrolled_at	TIMESTAMP	Required	Enrollment date
completed_at	TIMESTAMP	Nullable	Course completion date
created_at	TIMESTAMP	Auto	Record created
updated_at	TIMESTAMP	Auto	Record updated

Progress

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Progress ID
enrollment_id	BIGINT	FK → enrollments.id	Student's enrollment
lesson_id	BIGINT	FK → lessons.id	Lesson being tracked
status	ENUM	Not Started, In Progress, Completed	Lesson status
completed_at	TIMESTAMP	Nullable	Completion date
created_at	TIMESTAMP	Auto	Created
updated_at	TIMESTAMP	Auto	Updated

Notification

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Notification ID
user_id	BIGINT	FK → users.id	Recipient
title	VARCHAR(255)	Required	Notification title
message	TEXT	Required	Notification content
notification_type	ENUM	Required	Category of notification
is_read	BOOLEAN	Default false	Read status
read_at	TIMESTAMP	Nullable	When it was read
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Assessment_Attempt

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Attempt ID
enrollment_id	BIGINT	FK → enrollments.id	Student enrollment
assessment_id	BIGINT	FK → assessments.id	Assessment taken
attempt_number	INTEGER	Required	Attempt sequence (1, 2, 3...)
score	INTEGER	Required	Marks obtained
percentage	DECIMAL(5,2)	Required	Percentage at submission
is_passed	BOOLEAN	Required	Pass/fail result
started_at	TIMESTAMP	Required	Assessment start time
submitted_at	TIMESTAMP	Nullable	Submission time
created_at	TIMESTAMP	Auto	Created time
updated_at	TIMESTAMP	Auto	Updated time

Student_Answers

Column	Data Type	Constraints	Purpose
id	BIGINT	PK	Answer ID
attempt_id	BIGINT	FK → assessment_attempts.id	Parent attempt
question_id	BIGINT	FK → questions.id	Question answered
submitted_answer	TEXT	Required	Student's submitted answer
is_correct	BOOLEAN	Required	Correct or incorrect
marks_awarded	INTEGER	Default 0	Marks received
created_at	TIMESTAMP	Auto	Created
updated_at	TIMESTAMP	Auto	Updated

Module	Pattern	Confidence
User Management	Factory Method	⭐⭐⭐⭐⭐
Course Management	Builder	⭐⭐⭐⭐⭐
Enrollment Management	Strategy	⭐⭐⭐⭐⭐
Assessment & Progress	Observer	⭐⭐⭐⭐⭐
Notification	Command	⭐⭐⭐⭐☆

Module	Threat	OWASP Secure Coding Practice
User Management	Brute Force Login	**Enforce account disabling after an established number of invalid login attempts**
User Management	Password Disclosure	**If your application manages a credential store, use cryptographically strong one-way salted hashes**
Course Management	Unauthorized Course Modification	**Restrict access to protected functions to only authorized users**
Course Management	Stored XSS	**Contextually output encode all data returned to the client from untrusted sources**
Enrollment Management	Broken Access Control	**Enforce authorization controls on every request, including those made by server side scripts**
Enrollment Management	Privilege Escalation	**Use a single site-wide component to check access authorization. This includes libraries that call external authorization services**
Assessment & Progress	Information Disclosure	**Do not disclose sensitive information in error responses, including system details, session identifiers or account information**
Assessment & Progress	Lack of Audit Trail / Repudiation	**Log all access control failures.**
Notification Management	Man-in-the-Middle (MITM) Attack	**Consistently utilize HTTPS rather than switching between HTTP to HTTPS**
Notification Management	Insecure Communication with External Services	**Utilize TLS for connections to external systems that involve sensitive information or functions.**